
SDSC5001: Statistical Machine Learning I

Tutorial 2: Statistical Inference

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Tutorial Overview

This tutorial applies statistical inference ideas to new problems.

- ▶ Compare estimators using bias, variance, and MSE.
- ▶ Construct Method of Moments estimators.
- ▶ Apply maximum likelihood estimation.
- ▶ Investigate the effect of model assumptions.

Task 1: Bias, Variance, and MSE

The true parameter is $\theta = 10$.

Two estimators are evaluated on repeated samples.

	1	2	3	4
Estimator A	8	9	11	12
Estimator B	10.8	11.0	11.2	11.0

For each estimator:

1. Compute the average estimate.
2. Calculate the bias and variance.
3. Compute the mean squared error:

$$MSE = E[(\hat{\theta} - \theta)^2]$$

4. Verify:

$$MSE = Bias^2 + Variance$$

5. Decide which estimator has smaller overall error.

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Solution: Task 1 – Estimator A

Estimator A

The average estimate is:

$$\bar{\theta}_A = \frac{8 + 9 + 11 + 12}{4} = 10$$

Therefore, the bias is:

$$Bias_A = 10 - 10 = 0$$

The variance is:

$$\begin{aligned} Var_A &= \frac{(8 - 10)^2 + (9 - 10)^2 + (11 - 10)^2 + (12 - 10)^2}{4} \\ &= \frac{4 + 1 + 1 + 4}{4} = 2.5 \end{aligned}$$

Therefore:

$$Bias_A = 0, \quad Var_A = 2.5$$

Solution: Task 1 – Estimator B

Estimator B

The average estimate is:

$$\bar{\theta}_B = \frac{10.8 + 11.0 + 11.2 + 11.0}{4} = 11$$

Therefore, the bias is:

$$Bias_B = 11 - 10 = 1$$

The variance is:

$$\begin{aligned} Var_B &= \frac{(10.8 - 11)^2 + (11.0 - 11)^2 + (11.2 - 11)^2 + (11.0 - 11)^2}{4} \\ &= \frac{0.04 + 0 + 0.04 + 0}{4} = 0.02 \end{aligned}$$

Therefore:

$$Bias_B = 1, \quad Var_B = 0.02$$

Solution: Task 1 – MSE Decomposition

The mean squared error measures the overall estimation error:

$$MSE = E[(\hat{\theta} - \theta)^2]$$

Rewrite the estimation error:

$$\hat{\theta} - \theta = (\hat{\theta} - E[\hat{\theta}]) + (E[\hat{\theta}] - \theta)$$

Expanding:

$$(\hat{\theta} - \theta)^2 = (\hat{\theta} - E[\hat{\theta}])^2 + (E[\hat{\theta}] - \theta)^2 + 2(\hat{\theta} - E[\hat{\theta}])(E[\hat{\theta}] - \theta)$$

Taking expectation, the last term becomes zero.

Therefore:

$$MSE = \text{Variance} + \text{Bias}^2$$

Solution: Task 1 – Comparing Estimators

Using:

$$MSE = Bias^2 + Variance$$

Estimator A:

$$MSE_A = 0^2 + 2.5 = 2.5$$

Estimator B:

$$MSE_B = 1^2 + 0.02 = 1.02$$

	Bias	Variance	MSE
Estimator A	0	2.5	2.5
Estimator B	1	0.02	1.02

Estimator A is unbiased but has higher variance. Estimator B has some bias but smaller overall error.

Task 2: Method of Moments for a Uniform Model

Assume $X \sim \text{Uniform}(0, \theta)$, where θ is unknown.

The population mean is:

$$E[X] = \frac{\theta}{2}$$

Observed data:

1.5, 2.0, 3.5, 4.0

Answer the following questions:

1. Construct the Method of Moments estimator for θ .
2. Compute the estimate using the observed data.
3. Explain which population quantity is matched with which sample quantity.

Solution: Task 2 – Method of Moments for a Uniform Model

Method of Moments matches the population mean with the sample mean:

$$E[X] = \bar{X}$$

Using:

$$E[X] = \frac{\theta}{2}$$

we obtain:

$$\frac{\theta}{2} = \bar{X}$$

Therefore, the estimator is:

$$\hat{\theta}_{MM} = 2\bar{X}$$

Solution: Task 2 – Computing the Estimate

For the observed data:

$$1.5, \quad 2.0, \quad 3.5, \quad 4.0$$

the sample mean is:

$$\bar{X} = \frac{1.5 + 2.0 + 3.5 + 4.0}{4} = 2.75$$

Therefore:

$$\hat{\theta}_{MM} = 2(2.75) = 5.5$$

Solution: Task 2 – Interpretation

Method of Moments uses the following matching:

Population quantity	Sample quantity
$E[X] = \frac{\theta}{2}$	$\bar{X}$

The unknown parameter appears in the population quantity.

By replacing the population quantity with the observed sample quantity, we obtain an equation and solve for the parameter.

Main idea

Method of Moments connects what the model predicts with what the data show.

Task 3: Gaussian Maximum Likelihood

Assume:

$$X_i \sim \mathcal{N}(\mu, \sigma^2)$$

where μ and σ^2 are unknown parameters.

The observed data are:

$$1.2, \quad 1.8, \quad 2.4, \quad 3.0$$

Answer the following questions:

1. Compute the maximum likelihood estimate of μ .
2. Compute the maximum likelihood estimate of σ^2 .
3. Explain why the MLE variance estimator uses n instead of $n - 1$.

Solution: Task 3 – Estimating the Mean

For a Gaussian model, the maximum likelihood estimate of the mean is the sample mean:

$$\hat{\mu}_{MLE} = \bar{x}$$

For the observed data:

$$\bar{x} = \frac{1.2 + 1.8 + 2.4 + 3.0}{4} = 2.1$$

Therefore:

$$\boxed{\hat{\mu}_{MLE} = 2.1}$$

The MLE places the estimated mean at the center of the observed data.

Solution: Task 3 – Estimating the Variance

Using:

$$\hat{\mu}_{MLE} = 2.1$$

the squared deviations are:

$$(1.2 - 2.1)^2, \quad (1.8 - 2.1)^2, \quad (2.4 - 2.1)^2, \quad (3.0 - 2.1)^2$$

Their sum is:

$$0.81 + 0.09 + 0.09 + 0.81 = 1.8$$

The MLE variance is:

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2$$

Therefore:

$$\hat{\sigma}_{MLE}^2 = \frac{1.8}{4} = 0.45$$

Solution: Task 3 – Why n Instead of $n - 1$?

The maximum likelihood estimator is obtained by maximizing the likelihood function.
For the Gaussian model:

$$\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2$$

The denominator is n because this value is the direct solution of the likelihood optimization problem.

The commonly used sample variance:

$$s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$$

uses $n - 1$ to correct finite-sample bias.

MLE and unbiased estimation solve different problems: MLE maximizes likelihood, while $n - 1$ provides a bias correction.

Task 4: Likelihood Simulation

Generate Bernoulli observations with true probability $p = 0.7$.

Compare two sample sizes:

$$n = 20, \quad n = 200$$

For each sample:

1. Generate Bernoulli observations.
2. Compute the maximum likelihood estimate of p .
3. Evaluate the likelihood for candidate values of p from 0 to 1.
4. Plot the likelihood curve.
5. Compare the peak location and curve width.

Explain:

- Why do the two curves peak near similar values?
- Why is the larger sample curve narrower?

Solution: Task 4 – Likelihood Simulation

For Bernoulli observations, the maximum likelihood estimate is:

$$\hat{p}_{MLE} = \frac{\text{number of Heads}}{n}$$

For candidate values of p , compute:

$$L(p) = p^m (1 - p)^{n-m}$$

where m is the number of observed Heads.

The pseudocode is:

```
for n in [20, 200]:  
    generate n Bernoulli samples  
    m = number of Heads  
    p_hat = m / n  
    for p in grid(0, 1):  
        likelihood[p] = p**m * (1-p)**(n-m)  
    plot(p, likelihood)
```

Solution: Task 4 – Interpretation

Both likelihood curves should peak near:

$$\hat{p}_{MLE}$$

because the maximum likelihood estimate gives the parameter value that best explains the observed data.

The larger sample produces a narrower likelihood curve.

Main idea

The peak location gives the estimate. The width of the curve reflects how strongly the data support nearby parameter values.

Tutorial 2: calculate, compare, and understand estimation

Questions and discussion